Law of Sines and Law of Cosines

ANSWER KEY

Law of Sines

1 In triangle ABC, $A = 40°$, $B = 65°$, and $a = 12$. Find:

a angle C $= 75°$

b side b (2 d.p.) $= \frac{12 \sin 65°}{\sin 40°} \approx 16.92$

c side c (2 d.p.) $= \frac{12 \sin 75°}{\sin 40°} \approx 18.03$

2 Two observers 400 m apart on a straight road both sight a hot-air balloon between them. The angles of elevation are $58°$ and $47°$. Find the distance from the balloon to the first observer (the one with the $58°$ sighting), to the nearest metre.

The angle at the balloon is $180° - 58° - 47° = 75°$. By the Law of Sines, $d = \frac{400 \sin 47°}{\sin 75°} \approx 303$ m.

Law of Cosines

3 In triangle PQR, two sides of length 8 and 11 enclose an angle of $52°$. Find the length of the third side, to two decimal places.

$c^2 = 8^2 + 11^2 - 2(8)(11)\cos 52° \approx 185 - 108.36 = 76.64$, so $c \approx 8.75$.

4 A triangle has sides 7, 9, and 12. Find the measure of its largest angle, to the nearest tenth of a degree.

The largest angle is opposite the side of length 12: $\cos C = \frac{7^2 + 9^2 - 12^2}{2(7)(9)} = \frac{-14}{126} = -\frac{1}{9}$, so $C \approx 96.4°$.

5 Find the area of a triangle with sides 10 and 14 enclosing an angle of $35°$, to two decimal places.

$A = \frac{1}{2}(10)(14)\sin 35° \approx 70 \times 0.5736 \approx 40.15$ square units.

6 A ship sails 30 km due east, then turns and sails 45 km on a bearing of $N\,50°\,E$. Decide which law applies and find the ship's distance from its starting point, to the nearest kilometre.

The angle between the legs is $90° + 50° = 140°$. Law of Cosines:
$d^2 = 30^2 + 45^2 - 2(30)(45)\cos 140° \approx 2925 + 2068.4 = 4993.4$, so $d \approx 71$ km.

The ambiguous case (SSA)

7 In triangle ABC, $A = 30°$, $a = 5$, and $b = 9$. Determine how many triangles satisfy these conditions (zero, one, or two), and find all possible measures of angle B.

$\sin B = \frac{b \sin A}{a} = \frac{9(0.5)}{5} = 0.9$. This gives $B \approx 64.2°$ or $B \approx 115.8°$. Since $A + B < 180°$ in both cases, there are two possible triangles.

8 In triangle DEF, $D = 40°$, $d = 10$, and $e = 6$. Determine how many triangles satisfy these conditions, explaining your reasoning.

$\sin E = \frac{e \sin D}{d} = \frac{6 \sin 40°}{10} \approx 0.386$, giving $E \approx 22.7°$ or $E \approx 157.3°$. Since $d > e$ (the side opposite the given angle is longer), only the acute solution is valid: exactly one triangle exists.

Choosing the right law

9 For each triangle description, state whether the Law of Sines or the Law of Cosines is the appropriate first tool, and why:

a Two angles and one side are known (AAS). Law of Sines — a full angle-side ratio pair is available.

b Three sides are known (SSS). Law of Cosines — no angle is known to pair with a side.

c Two sides and the included angle are known (SAS). Law of Cosines — the included angle lets you find the third side directly.

10 A triangular plot of land has sides 120 m, 150 m, and 90 m. Find its area, to the nearest square metre, using the fact that you can first find any angle with the Law of Cosines and then use $A = \frac{1}{2}ab\sin C$.

Find angle C between the sides of length 120 and 150 (opposite the side 90): $\cos C = \frac{120^2 + 150^2 - 90^2}{2(120)(150)} = \frac{28800}{36000} = 0.8$, so $C \approx 36.87°$. Area $= \frac{1}{2}(120)(150)\sin(36.87°) \approx 9000(0.6) = 5400 \text{ m}^2$.

Heron's formula

11 State Heron's formula for the area of a triangle with sides a, b, c and semi-perimeter $s = \frac{a+b+c}{2}$, then use it to find the area of a triangle with sides 13, 14, 15.

$A = \sqrt{s(s-a)(s-b)(s-c)}$. Here $s = \frac{13+14+15}{2} = 21$. $A = \sqrt{21(8)(7)(6)} = \sqrt{7056} = 84$.

12 A triangular garden has sides 9 m, 12 m, and 15 m. Use Heron's formula to find its area, and verify it is a right triangle by checking the Pythagorean relationship.

$s = \frac{9+12+15}{2} = 18$. $A = \sqrt{18(9)(6)(3)} = \sqrt{2916} = 54 \text{ m}^2$. Check: $9^2 + 12^2 = 81 + 144 = 225 = 15^2$ ✓ — confirms a right triangle, and indeed $\frac{1}{2}(9)(12) = 54$ matches.

Navigation and bearing problems

13 A plane flies on a bearing of $N\,60°\,E$ for 200 km, then changes course to a bearing of $S\,80°\,E$ for 150 km. Find the angle between the two legs of the flight, and set up (but do not evaluate) the Law of Cosines expression for the total displacement.

The angle between the two bearings: turning from $N60°E$ to $S80°E$ sweeps through $180° - 60° - 80° = 40°$ measured one way, but the interior angle of the path (between the two flight segments) is $180° - 40° = 140°$. Displacement: $d^2 = 200^2 + 150^2 - 2(200)(150)\cos(140°)$.

Solving oblique triangles from a diagram

14 In triangle XYZ, $x = 8$, $y = 10$, and angle $Z = 48°$ (between the given sides). Find side z, then find angle X using the Law of Sines, to the nearest tenth of a degree.

$z^2 = 8^2 + 10^2 - 2(8)(10)\cos 48° \approx 164 - 107.0 = 57.0$, so $z \approx 7.55$. Then $\sin X = \frac{8\sin 48°}{7.55} \approx 0.787$, giving $X \approx 51.9°$.

Verifying triangle solutions with the angle sum

15 For the triangle solved above ($z \approx 7.55$, $X \approx 51.9°$, $Z = 48°$), find angle Y and verify all three angles sum to $180°$.

$Y = 180° - 48° - 51.9° = 80.1°$. Check: $48° + 51.9° + 80.1° = 180°$ ✓.

Choosing between multiple valid approaches

16 A triangle has $a = 6$, $b = 8$, $c = 10$. Find angle C (opposite the longest side) two different ways: (1) using the Law of Cosines, and (2) recognizing this as a Pythagorean triple.

(1) $\cos C = \frac{6^2 + 8^2 - 10^2}{2(6)(8)} = \frac{0}{96} = 0 \Rightarrow C = 90°$. (2) Since $6^2 + 8^2 = 36 + 64 = 100 = 10^2$, this is a right triangle by the Pythagorean theorem, confirming $C = 90°$ directly — both methods agree.